

Juan Manuel Young Hoyos

SENIOR PYTHON BACKEND ENGINEER | NESTJS | FASTAPI | MICROSERVICES | HL7 FHIR

Medellín, Colombia · English B2+ (professional working proficiency)

☎ +57 1234567890 | ✉ juanmanuel12.13jmyh81@gmail.com | 🌐 jmyoung hoyos.com | 📺 Youngermaster | 📄 juan-manuel-young-hoyos

Summary

Backend-focused senior engineer with 5+ years building production software, using **Python** across services, automation and tooling since 2021, alongside **NestJS**. I design **RESTful APIs** and **microservices** with **FastAPI** and **async/await**, queue-driven architectures with **background workers** for long-running jobs, model **MongoDB** persistence for production workloads, and secure them with **OAuth 2.0 / OIDC / JWT**. Currently leading a national **healthcare interoperability** platform built on **HL7 FHIR R4**. I ship with **Docker**, **Pytest**, **CI/CD** and observability, and use **AI-assisted development** (Claude Code, LLMs) while reviewing and validating every generated line.

Skills

Python & Backend	Python, FastAPI, async/await, RESTful API design, microservices, distributed systems, NestJS/Node.js, Go
Data & Messaging	MongoDB (production modelling), Redis, RabbitMQ, Kafka, Beanie ODM (familiar)
Auth & Security	OAuth 2.0, OpenID Connect (OIDC), JWT, identity providers; DevSecOps (OWASP ZAP, SonarQube)
Architecture & Performance	Software architecture, scalability, reliability, performance optimization, troubleshooting
Testing & Quality	Pytest, Jest, Playwright, Selenium, unit, integration & end-to-end tests, code reviews, debugging
Cloud & DevOps	Docker & Docker Compose, AWS, Azure, CI/CD (GitHub Actions, GitLab), observability (Grafana, Loki)
AI-Assisted Development	Claude Code and LLMs for code generation, refactoring, test generation and documentation
Healthcare Interop	HL7 FHIR R4, integrations with hospital & government systems, national compliance pipelines

Experience

Okorum Technologies

Medellín, Colombia (Hybrid)

TECHNICAL LEAD

Dec. 2025 – Present

- Delivered a national **healthcare interoperability** platform **ahead of the government mandate deadline** by architecting distributed **microservices** that stream send / validate / convert / generate stages over **message queues** into standards-compliant **HL7 FHIR R4** resources for EHR data exchange.
- Gave providers operational analytics by building dedicated **Python FastAPI** microservices, keeping API responses fast by offloading **long-running aggregation jobs** to **asynchronous background workers**.
- Kept the pipeline fast under sustained multi-provider load by modelling **MongoDB** persistence, adding a **Redis** caching layer, and engineering **fault-tolerant, resumable asynchronous queues** that recover from an unreliable upstream government API.
- Led a cross-functional **Agile** team through **architecture reviews**, **code reviews** and release while **mentoring** engineers on backend practices, and contributed platform **observability** (Grafana, Loki) with **CI/CD** on AWS (GitHub Actions).

Publicis Global Delivery (PGD) — Digitas / Razorfish

Colombia (Remote)

FRONTEND / REACT NATIVE DEVELOPER

Dec. 2023 – Nov. 2025

- Built and shipped **internal tooling** used across the agency — **Unicorn AI**, an **AI-agent platform** automating workflows for developers, designers, accountants and lawyers — contributing to its **Python RAG** backend alongside its React / Nuxt.js front ends.
- Cut critical bugs by **25%** with **automated testing** and Dynatrace monitoring, and reduced release-cycle time by **50%** by automating **CI/CD** pipelines with GitHub Actions.

Grisú

Colombia (Remote)

FULL STACK DEVELOPER (SELF-EMPLOYED)

Jul. 2023 – Present

- Compressed a 4-month plan into weeks by auditing and accelerating **EcoDrive**, an **event-driven** system built with **Python (FastAPI)** and **Kafka**.
- Delivered services to **Azure** with secured **CI/CD**, implementing **JWT / OAuth 2.0** authentication for stakeholder-facing applications.

Design Systems Inno

Medellín, Colombia (Hybrid)

BACKEND DEVELOPER / PROBLEM SOLVER

Dec. 2022 – Jun. 2023

- Removed manual back-office work for **hospitals** nationwide by building **Python RPA** automations that drive **government institution** portals end to end — **Playwright** browser automation with BeautifulSoup HTML parsers — plus an automated contract-signing pipeline.
- Built production **REST APIs in Python (FastAPI)** and Go (Go Fiber) on **MongoDB**, kept responsive under heavy automation load by a decoupled **asynchronous** processing layer on **RabbitMQ** where clients poll **long-running jobs** by ticket ID instead of blocking.
- Sped up delivery of **containerized** services by improving **Docker** image quality and build times, and paired with teammates on **root-cause analysis** of critical production bugs to keep releases on schedule.

ArcDesign • Geta Club Play & Aligo

Spain / Colombia (Remote & On-site)

EARLIER EXPERIENCE — PYTHON AUTOMATION & SECURITY ENGINEERING

Sep. 2021 – Mar. 2025

- Wrote **Python** simulation, automation and security tooling; reduced known vulnerabilities **50%** in 4 months via **DevSecOps** pipelines.

Education

EAFIT University

Medellín, Colombia

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Jan. 2018 – Dec. 2025

- Full scholarship “Ser Pilo Paga” (national merit-based award).